

CS6P008 - Deep Learning Lab
Lab 6, Date: March 16, 2026
Topic(s): Weight Initialization Strategies

Timing: 10:00 AM to 01:00 PM

Max Points: 10

Instructions

1. Provide commented, indented code. Variables should have meaningful names.
2. Submit two .ipynb files.
The names should be `[student name]_lab[number]_1.ipynb` (before completion of the lab) and `[student name]_lab[number]_2.ipynb` (your final code file).
3. Read the questions carefully before answering. If a question asks to follow a particular approach or to use a specific data structure, then it must be followed.
4. Write questions in separate text blocks in Jupyter/Colab Notebook before the code blocks containing answers.
5. All plots should have appropriate axis labels, titles, and legends.

Do not use the solutions provided by third-party libraries such as scikit-learn.
You can use `numpy`, `pandas`, `matplotlib`.

1. Implement a 5-layer neural network for MNIST classification with the following specifications:
 - (a) Apply standardization and normalization to the dataset if necessary.
 - (b) Create a fully connected neural network with 5 hidden layers.
 - (c) Use the following activation functions `Relu`, `Sigmoid`, `Tanh`, `LeakyRelu`, `Exponential Linear Unit` (ELU) respectively after each hidden layer.
 - (d) Apply **Batch normalization** after each layer.
 - (e) Initialize the weights of the network using:
 - **All Zero Initialization**
 - **Uniform Initialization**
 - **Standard Normal Initialization**
 - **LeCun Uniform Initialization**
 - **Xavier Uniform Initialization**
2. Perform the following:
 - (a) Perform a forward pass through the network.
 - (b) Capture the output activations after each hidden layer.
 - (c) Plot histograms of the activations after each layer. Refer to Figure 1 for sample expected output.
 - (d) Plot histograms of the gradients after each layer. Refer to Figure 2 for sample expected output.
 - (e) Repeat this for all weight initialization methods mentioned above.

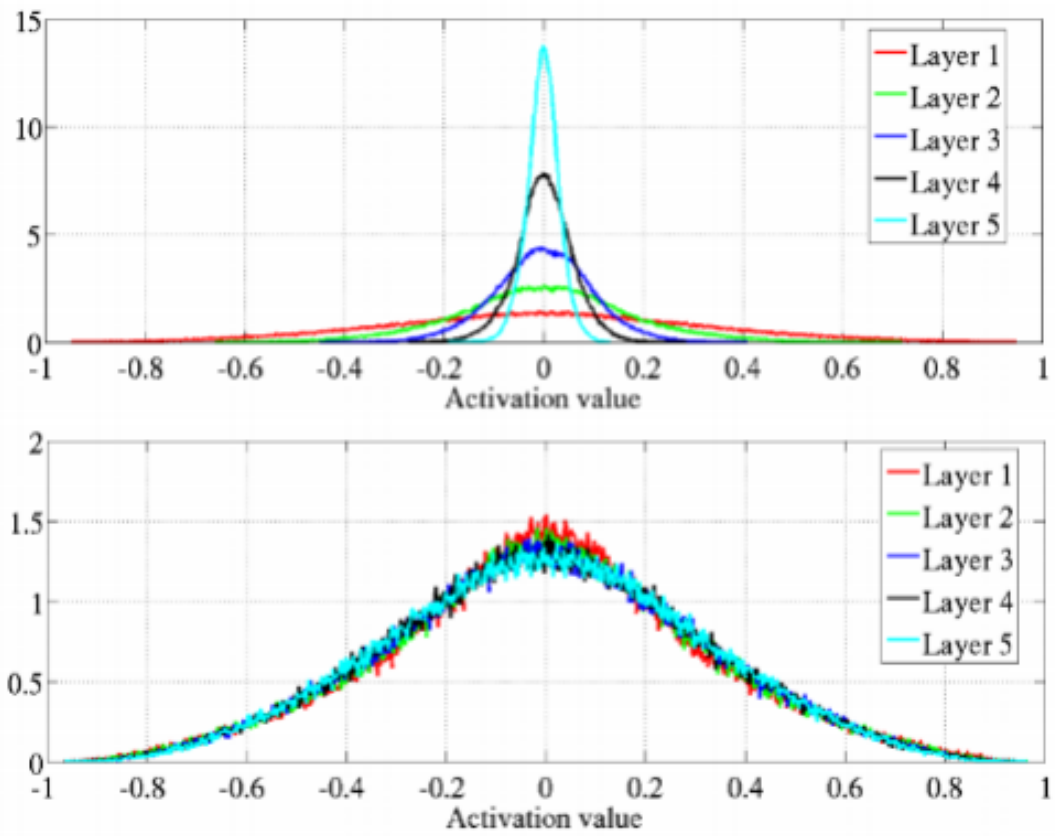


Figure 1: First row: For random initializations, Second Row: For Xavier initializations

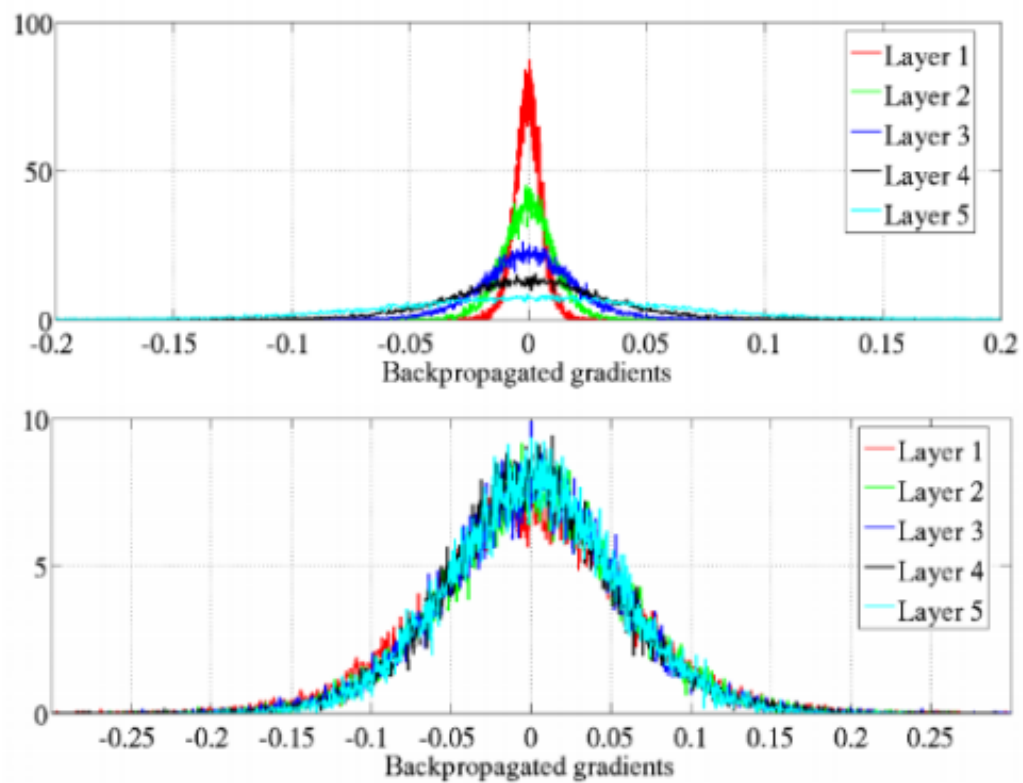


Figure 2: First row: For random initializations, Second Row: For Xavier initializations